

Vu Phan

Hardware & Firmware Engineer

Ho Chi Minh City, Vietnam

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Professional Summary

Hardware and firmware engineer with over 16 years of experience delivering end-to-end product development for IoT, consumer electronics, telecom, and automotive industries. Proficient in MCU, SoC, and FPGA-based system design, high-speed interfaces (PCIe, DDR, Ethernet), and embedded protocol stacks (UART, SPI, I2C). Across the full stack - from schematic and layout to firmware and application software - I build reliable, production-ready systems.

Skills

Hardware Design MCU/SoC/FPGA system design, PCB layout, high-speed multi-layer board design

Protocols & Interfaces PCIe, DDR, Ethernet, UART, SPI, I2C

Programming C, C++, C#, JavaScript, Dart/Flutter, Python, Node.js

EDA & CAD OrCAD Capture, Concept HDL, Allegro, Altium Designer, EAGLE, KiCad

Mechanical FreeCAD, SolidWorks

Experience

Cadera Systems - Hardware & Firmware Engineer

Jul 2011 - Present

Hardware Design

- Architected and delivered custom hardware for IoT, consumer electronics, telecom, and automotive clients, from specification through schematic design, layout, and prototyping
- Led board bring-up and validation, debugging complex signal-integrity and power-delivery issues on high-speed multi-layer PCBs

Embedded Firmware Development

- Developed bare-metal and RTOS-based firmware for MCU, SoC, and FPGA platforms, implementing peripheral drivers, communication stacks, and bootloaders

Software & Application Development

- Built cross-platform test tools, data-logging applications, and host-side utilities in C#, Dart/Flutter, and Node.js to support hardware validation and production

Acronics Systems, Inc - Firmware / Software Developer

Jan 2010 - Jul 2011

- Developed Linux device drivers and application software for Set-Top-Box platforms, working across the kernel-to-userspace boundary

Education

HCMC University Of Technology

Bachelor of Engineering in Electronics

2005 - 2010

Languages

Vietnamese Native

English Professional proficiency